

Joy Mondal

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Education

VIT Bhopal University

B.Tech in Computer Science and Engineering (Cyber Security & Digital Forensics), CGPA: 7.94/10

Bhopal, Madhya Pradesh

2022 – 2026

Projects

Watchr – Movie & TV Tracker – React, FastAPI, PostgreSQL, Redis, Celery, Docker, WebSockets

Jul 2026

- Designed and developed a full-stack movie and TV tracking platform featuring JWT authentication, watchlists, ratings, viewing history, real-time activity updates, and offline-first synchronization.
- Built an asynchronous synchronization pipeline using Celery and Redis with retry policies, idempotent workers, exponential backoff, and rate-limited TMDB API integration, reducing metadata retrieval latency by over 80% through Redis caching.

Urldis – Distributed URL Shortener – FastAPI, PostgreSQL, Redis, RabbitMQ, Prometheus, Grafana

Jun 2026

- Designed and developed a distributed URL shortening platform implementing Redis cache-aside caching, negative caching, collision-safe Base62 short-code generation, and low-latency URL redirection.
- Engineered an event-driven backend using RabbitMQ and Celery for asynchronous analytics processing, integrated Prometheus and Grafana for observability, and structured the application using Clean Architecture (DDD).

SOC Home Lab – Wazuh, Sysmon, Suricata, Zeek, Elasticsearch, Kibana, Docker

Feb 2026 – May 2026

- Built a Security Operations Center (SOC) home lab integrating endpoint and network security tools to centralize, correlate, and visualize 200–500+ security events for monitoring and analysis.
- Conducted 20+ MITRE ATT&CK-based attack simulations, developed 15+ custom detection rules, investigated security alerts through Kibana dashboards, and validated threat detection and incident response workflows.

Toxicity Prediction using GAN – Python, PyTorch, PyTorch Geometric, RDKit, NumPy, Matplotlib

Feb 2026

- Developed a multi-task molecular toxicity prediction system using Graph Attention Networks (GATs) on the NIH Tox21 dataset containing 8,000+ compounds across 12 toxicity endpoints.
- Implemented SMILES-to-graph conversion, multi-head graph attention, ROC-AUC evaluation, and model interpretability techniques including attention visualization and molecular feature importance analysis.

Activities

Cyber Warrior VITB – Sr. Tech Lead

Sep 2023 - Apr 2024

- Coordinated 8+ cybersecurity workshops, technical events, and hands-on sessions involving 100+ participants, ensuring smooth execution and active engagement.
- Designed and developed 15+ Capture The Flag (CTF) challenges spanning web security, cryptography, forensics, and reverse engineering for competitive learning environments.

Microsoft Technical Club VITB – Jr. Tech Lead

Nov 2023 - Aug 2024

- Organized and executed 10+ technical and non-technical events, including coding competitions, workshops, and interactive learning sessions with 200+ attendees.
- Managed event operations during AdvITya, the university's annual fest, coordinating cross-functional teams to ensure seamless execution and high participant engagement.

Achievements

Smart India Hackathon 2024 – Qualified in Internal Round, VIT Bhopal University

May 2024

- Developed a solution in a 6-member team, for a problem statement with 500+ national submissions.

SecVit'24 – Runner UP, Null VIT Bhopal Chapter, VIT Bhopal

Jun 2024

- Solved 10+ CTFs, outperforming 100+ participants to earn certification.

BugTrek'24 – 3rd Place, Null VIT Bhopal Chapter, VIT Bhopal University

Dec 2024

- Solved 10+ CTFs, outperforming 125+ participants to earn certification.

Certifications

Google Cloud Arcade – Google Cloud

May 2025

Blockchain Developer – IBM

Apr 2025

Cryptography I – Stanford Online

Feb 2025

The Bits and Bytes of Computer Networking – Google, Coursera

Dec 2023

Skills

Programming Languages: Python, Java, JavaScript, SQL, Bash, HTML5, CSS3

Frameworks & Libraries: FastAPI, React.js, Tailwind CSS, Vite, SQLAlchemy, PyTorch, PyTorch Geometric, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Databases & Messaging: PostgreSQL, SQLite, Redis, RabbitMQ

DevOps & Observability: Docker, Git, GitHub, GitHub Actions, Prometheus, Grafana, Celery, Linux, Postman

Cybersecurity: Wazuh, Sysmon, Suricata, Zeek, Elasticsearch, Kibana, MITRE ATT&CK, Threat Hunting, Incident Response

Machine Learning: CatBoost, Graph Neural Networks (GAT), Feature Engineering, Model Evaluation, Data Visualization

Core Concepts: Distributed Systems, Event-Driven Architecture, Clean Architecture (DDD), REST APIs, WebSockets, JWT Authentication, Asynchronous Processing, Caching, Message Queues, System Design

Operating Systems: Linux (Ubuntu, Fedora, Arch, Kali), Windows